

# MINJAE KWON

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## Research Interests

**Engineering nanostructures that shape light and fluidic behavior, guided by computational modeling.**

**Nanofabrication:** periodic/3D nanostructures with tailored optical, mechanical, and interfacial properties.

**Plasmonics & Metasurfaces:** metasurface engineering for plasmonic field enhancement and optical sensing.

**Electrokinetics & Simulations:** multiphysics FEA optimizing electrokinetic particle capture and optical enhancement.

**Deep-Learning Readout:** CNN-based image analysis for quantitative particle-capture recognition.

## Education

**Korea University** (Seoul, Republic of Korea) Sep. 2025 – Present

*Master's Student in Department of Bioengineering*

- Advised by Prof. Yong-Sang Ryu

**Korea University** (Seoul, Republic of Korea) Mar. 2023 – Aug. 2025

*B.S. in Biomedical Engineering & Electrical Engineering (Double Major)*

- GPA : N.A / 4.50

**Konkuk University** (Seoul, Republic of Korea) Mar. 2017 – Feb. 2023

*Majored in Electronic and Electrical Engineering*

- Transferred to Korea University
- Leave of absence due to national military service: Jul. 2019 – Jan. 2021

## Research Experience

**Ryu group, Korea University** (Seoul, Republic of Korea) Aug. 2024 – Present

*Undergraduate/Graduate Student Researcher (Advised by Prof. Yong-Sang Ryu)*

- **Current work: MIM Nanohole Plasmonic Metasurface for Single-Particle Analysis**  
Fabrication: Fabricating large-area nanohole structures via laser interference lithography (LIL), engineering a single-nanoparticle-capture geometry with a metal–insulator–metal dielectric stack to amplify SERS enhancement beyond prior work, enabling electrokinetic trapping of single particles at the hotspots.  
Simulation: Modeling optical (COMSOL ewfd) and electrokinetic (COMSOL ec/spf) fields to find geometries giving up to  $\sim 1.8 \times 10^5$ -fold field enhancement, and validating DEP/ACEO single-particle capture dynamics.  
CNN Analysis: Applying CNNs to SEM images to quantify the relative standard deviation (RSD) of single-nanoparticle capture to analyse biosensor uniformity and the resulting SERS signal.
- **Collaborative projects**  
PRET metasurface: Depositing capture structures onto the Plasmonic Resonance Energy Transfer metasurface built by Prof. Junsuk Noh's group, enabling electrokinetic concentration of target analytes in aqueous solution.  
Hydrogen-sensing MEMS: Providing deposition and lithography support for a terahertz-resonant MEMS chip whose resonance shift enables label-free hydrogen sensing.  
Electrochemical-sensing nanostructure: Leading full-chip fabrication of a capture-free MIM nanohole variant — tuning hole size, nano-gap thickness, and materials for fast and sensitive electrochemical detection.  
KU–KIST joint project: Serving as a collaborating researcher at the Nanophotonics Research Center, KIST — developing label-free plasmonic metasurface for sensing GPCR conformational dynamics.

**Creative Challenger Program, Korea University** Aug. 2023 – Feb. 2024

*Project Leader, Vine Robot Colonoscopy Project (Advised by Prof. Daehee Hong and Dr. Sang-Hyun Kim)*

- Developed and validated a novel soft-robot colonoscopy platform; conducted in-vivo experiments in collaboration with the Department of Gastroenterology, Korea University Medical Center.
- Recognized with two Excellence Awards (15<sup>th</sup> Creative Challenger Program, 2<sup>nd</sup> Changemakers Forum)

## Technical Proficiency

### Fabrication & Characterization Skills

- Lithography: Photolithography, Laser Interference Lithography (LIL), Electron Beam Lithography (EBL), Soft Lithography, Nanoimprint Lithography (NIL)
- PVD (Physical Vapor Deposition): Thermal Deposition, Electron Beam Deposition, Sputter Deposition
- CVD (Chemical Vapor Deposition): ALD, PECVD
- Etching: RIE, ICP-RIE
- Surface Treatment: Oxygen Plasma Ashing, UV-Ozone
- Material / Structure Analysis: SEM, AFM, Energy Dispersive X-ray Spectroscopy (EDS), Surface Profiler, Ellipsometry, Reflectance Spectroscopy, Raman Spectroscopy, Optical & Epi-fluorescence Microscope

### Computational & Design Skills

- Language: Python, C/C++, LaTeX, MATLAB
- FEA Simulation: COMSOL Multiphysics, Lumerical FDTD
- Data Analysis: OriginLab, Veusz, Python-based data analysis (PyTorch, OpenCV)
- 2D/3D Design: Rhino, AutoCAD, Keyshot
- Physical Design: PCB/Breadboard Circuit Design, 3D Printing, Metal / Polymer CNC, Laser Cutting

## Publications

[1] S. Lee<sup>‡</sup>, S. Choi<sup>‡</sup>, T. Kim, **M. Kwon**, T. J. Mun, S. Son, H. Seong, K. Lee, H. Im, E.-S. Yu, M.-K. Kim<sup>\*</sup>, and Y.-S. Ryu<sup>\*</sup>, "*Analyte-accessible field volume as a governing design principle for nanophotonic sensing*," under review, Advanced Materials (2026).

<sup>‡</sup>Equal contribution, <sup>\*</sup>Corresponding author

## Leadership & Extracurricular Activities (Selected)

### School of Biomedical Engineering, Korea University

*Founder & President, KU Biomedical Eng. Abroad Mentorship (KUBEAM)* Aug. 2025 – Present

- Founded a mentorship initiative connecting 80+ students and alumni pursuing study/careers abroad, building a shared study-abroad resource hub and overseas alumni network.

*Biosensor Team Lead Organizer, Biomedical Engineering Academic Team (BEAT)* Aug. 2025 – Jul. 2026

- Led and mentored 60+ undergraduates in acquiring the concepts and technical skills necessary to explore the field of biosensors (e.g., SEM imaging, data analysis, critical literature review, presentation).

*Senior Advisor & Vice President, Department Student Council* Nov. 2024 – Nov. 2025

- Advised and supported planning and execution of academic, cultural, and community events for 500+ undergraduates. Led a 30+ member committee and secured \$22,000 in sponsorships for a 500+ student alumni day.

### 65<sup>th</sup> Medical Brigade, 8<sup>th</sup> U.S. Army

Jul. 2019 – Jan. 2021

*Combat Medic & Medical Support Assistant, BDAACH (549<sup>th</sup> Hospital Center)*

- Selected for the KATUSA (Korean Augmentation to the U.S. Army) program based on language proficiency. Served as staff in the Physical Performance Service Line (musculoskeletal rehabilitation and physical therapy) for outpatient care and on the COVID-19 Response Team (quarantine operations, vaccine administration).
- Served as Senior KATUSA of 65<sup>th</sup> Medical Brigade, the ranking KATUSA representative and liaison to U.S. command; received the Army Achievement Medal; honorably discharged as Sergeant (E-5).

## Teaching Experience

### Teaching Assistant, Guest Lecturer

Biomedical Semiconductor Process (BMED442, Prof. Yong-Sang Ryu)

Fall 2025, Fall 2026

Biomedical Design (HBL514, Prof. Yong-Sang Ryu)

Spring 2026

## Awards and Scholarships (Selected)

**BK21 FOUR Graduate Research Scholarship**, Ministry of Education (Full Tuition)

Sep. 2025 – Present

**Best Poster Presentation Award**, Korean Biochip Fall Symposium

Nov. 2025

**Outstanding New Student Scholarships**, Korea University

Sep. 2025

**Dean's List (Highest Honors, 2 semesters)**, Korea University

Mar. 2023 – Jun. 2025

**Excellence Award**, 2<sup>nd</sup> Changemakers Forum

Nov. 2024

**Best Startup Award**, Korea University

Nov. 2024

**Third Place**, Capstone Design Fair

Sep. 2024

**Excellence Award**, 15<sup>th</sup> Creative Challenger Program

Feb. 2024

## Languages

**Korean**

Native speaker

**English**

TOEFL 5.0 / 6.0 (equivalent to 100)